

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CE/EE/IT) Sem - I Examination, 2009-10

- CL - 103 Mechanics of Solids

Date: 18.12.09, Friday

Time: 02.00 p.m to 05.00 p.m

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION- I

- Q. 1 (a) State & Prove law of parallelogram. 03
- (b) State Pappus-Guldinus 1st and 2nd theorems. 04
- Q. 2 (a) A cord supported at A and B carries a load of 10 kN at D and a load W at C as shown in figure 1. Find the value of W so that CD remains horizontal. 04
- (b) Find resultant for the force system shown in figure 2 with respect to point A. 06
- (c) Locate centroid of the plane area shown in figure 3. 04

OR

- Q. 2 (a) Two smooth spheres each of radius 100mm and weight 100 N rest in a horizontal channel having vertical walls, the distance between which is 360 mm. Find reactions at points of contacts A, B, C and D. shown in figure 4. 04
- (b) Find the magnitude of force F_2 and angle θ for the figure 5. 06
- (c) Locate centroid for the bent ABCDEFGH with respect to the axis shown in figure 6. 04
- Q. 3 (a) A ladder is 6m long and is supported by a horizontal floor (A) and a vertical wall (B). The coefficient of friction between the floor and ladder is 0.25 and between the wall and the ladder is 0.4. The weight of the ladder is 200 N and acting at the middle. The ladder supports a vertical load of 900 N at a distance 1m from top i.e. from B. Determine the least value of inclination of ladder with horizontal (θ) which the ladder may be placed without slipping. 07
- (b) Find forces in all members of the cantilever truss shown in figure 7. 07

OR

- Q. 3 (a) Block A, weighing 10 kN is to be raised against a vertical wall as shown in figure 8 by means of 15° wedge B. Find the smallest horizontal force P required to raise the block A. Take coefficient of friction 0.2 at all contact surfaces. 07

- (b) Determine the forces in members FH, GH and GI for the truss shown in figure 9. 07

SECTION- II

- Q. 4 (a) Define Hardness, Malleability, Ductility and Poisson's Ratio. 03

- (b) Explain stress-strain curve for ductile material with salient points. 04

- Q. 5 (a) Find total deformation of the member shown in figure 10. Take $E = 2 \times 10^5 \text{ N/mm}^2$. 07

- (b) A specimen of steel 20mm diameter with a gauge length of 200mm is tested to destruction. It has an extension of 0.25mm under a load of 80 kN and the load at elastic limit is 102 kN. The maximum load is 130 kN. 07

The total extension at fracture is 56mm and diameter at neck is 15mm. Find

- i. Stress at elastic limit
- ii. Young's modulus
- iii. Percentage elongation
- iv. Percentage reduction in area
- iii. Ultimate tensile stress

OR

- Q. 5 (a) A reinforced concrete column of size 300mm x 500mm has 8 steel bars of 16mm diameter. If the column is subjected to an axial compressive force of 800 kN, find the stresses developed in steel and concrete. Take $E_s/E_c = 18$. 07

- (b) A steel rail is 12m long and is laid at a temperature of 18°C . The maximum temperature expected is 40°C . Take $\alpha = 12 \times 10^{-6}/^\circ \text{C}$ and $E = 2 \times 10^5 \text{ N/mm}^2$. 07

- i. Estimate the minimum gap between two rails to be left so that temperature stresses do not develop.

- ii. Calculate the temperature stresses developed in the rails if:

a) No expansion joint is provided.

b) If a 1.5 mm gap is provided for expansion.

- Q. 6 (a) Define shear force, bending moment and point of contra-flexure. 04

- (b) Draw shear force and bending moment diagrams indicating values at all important points for an overhanging beam shown in figure 11. Also locate point of contra-flexure if any. 10

OR

- Q. 6 (a) Derive relation between load, shear force and bending moment with usual notations. 04

- (b) Draw shear force and bending moment diagrams indicating values at all important points for a simply supported beam shown in figure 12. 10

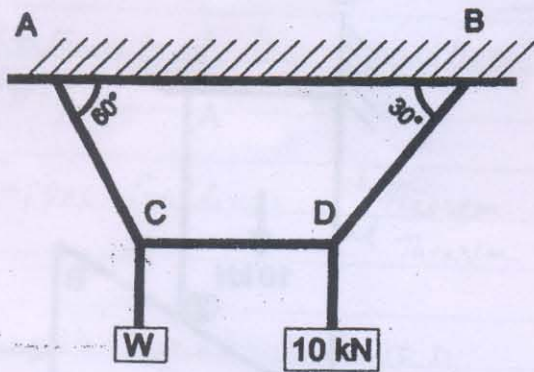


Figure 1

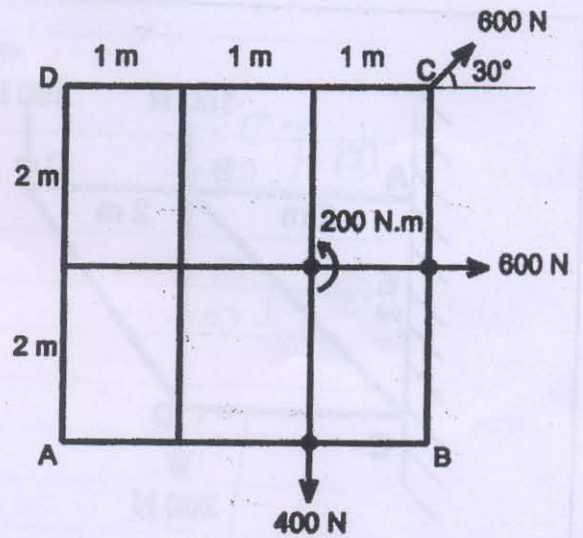


Figure 2

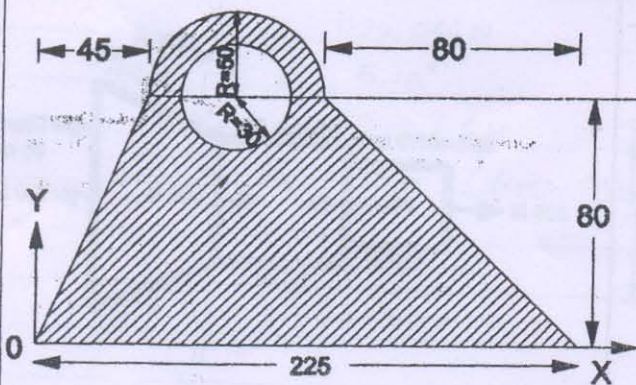


Figure 3

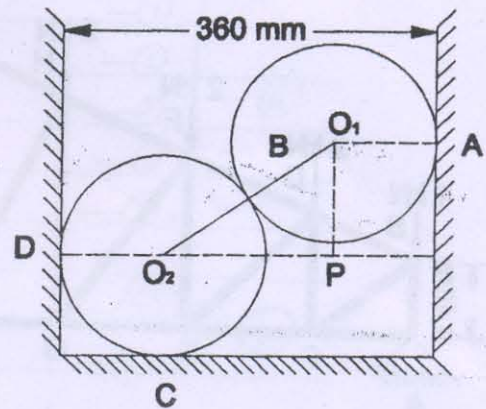


Figure 4

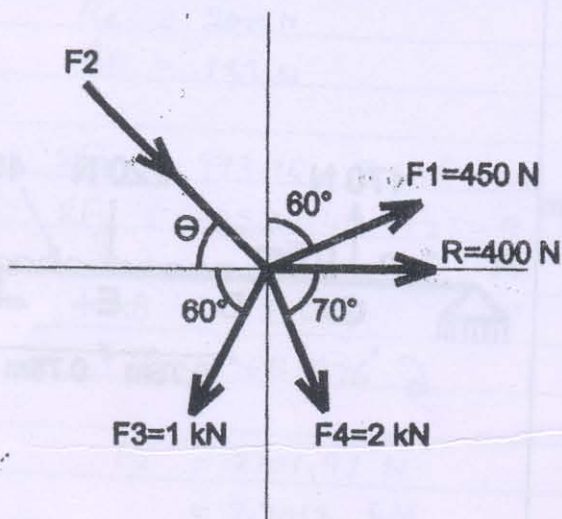


Figure 5

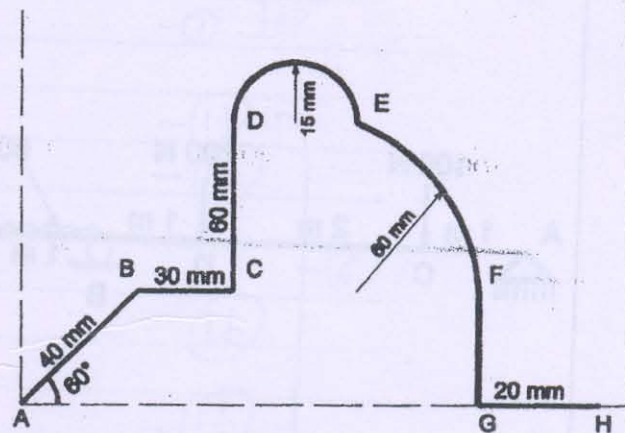


Figure 6

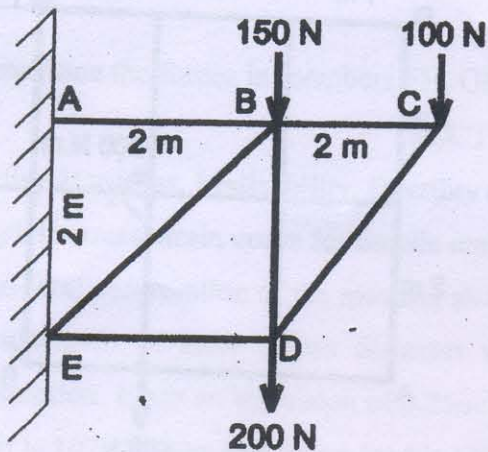


Figure 7

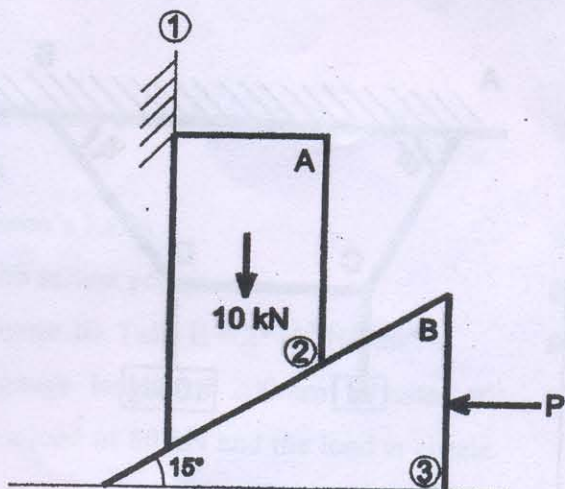


Figure 8

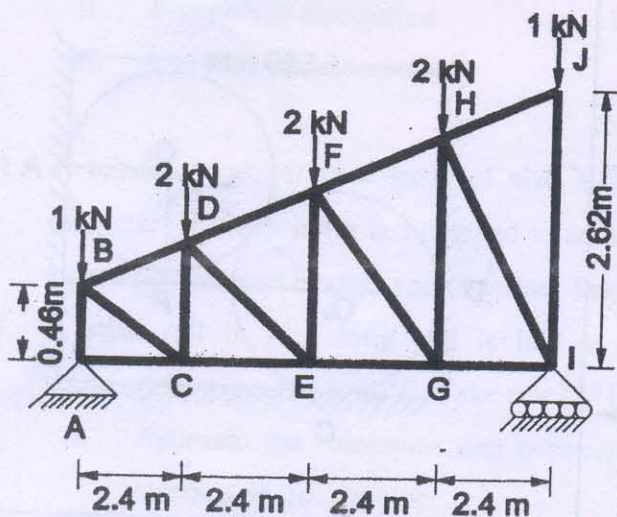


Figure 9

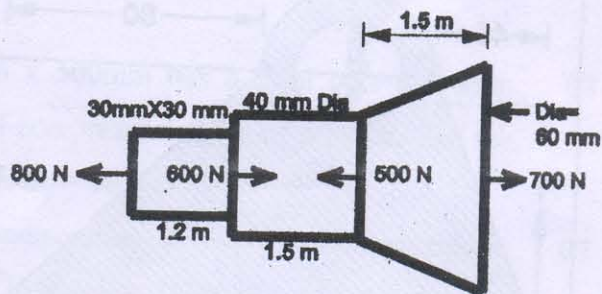


Figure 10

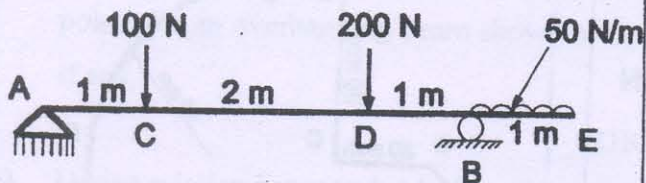


Figure 11

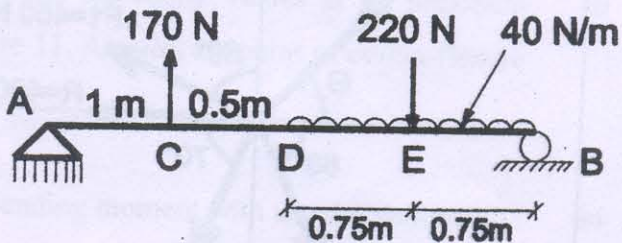


Figure 12